



Class 9 2022-23

FORCE AND LAWS OF MOTION

Ch 9 | Class 9th (Science)

Handwritten Notes & Imp Questions



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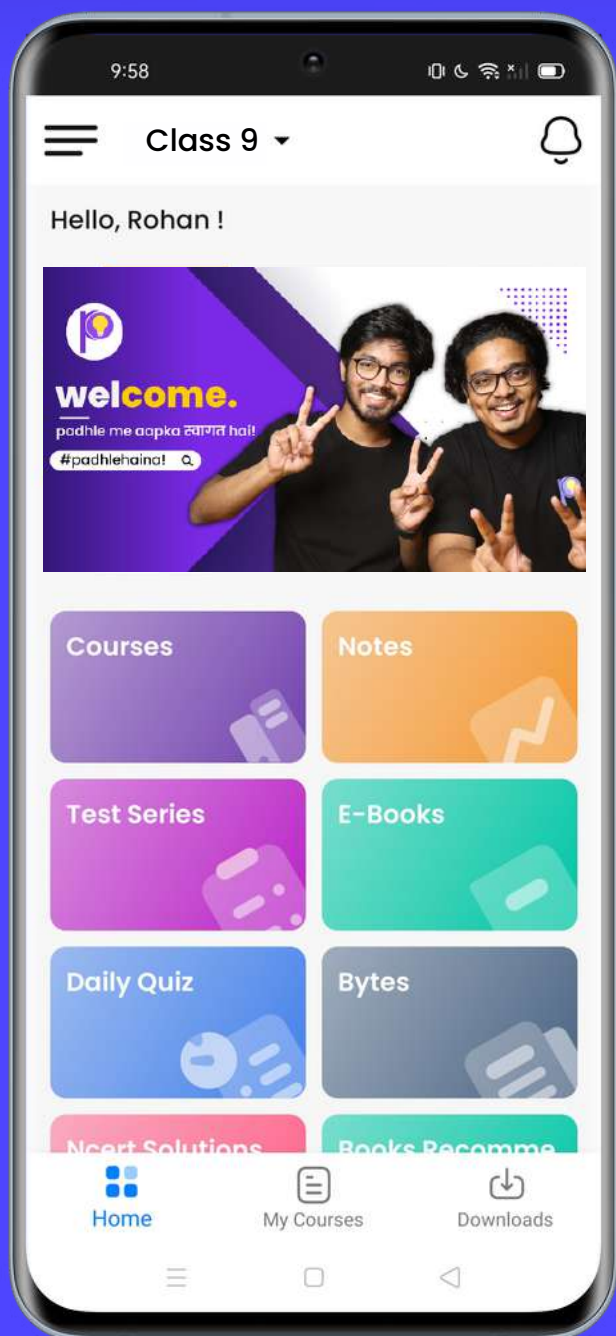
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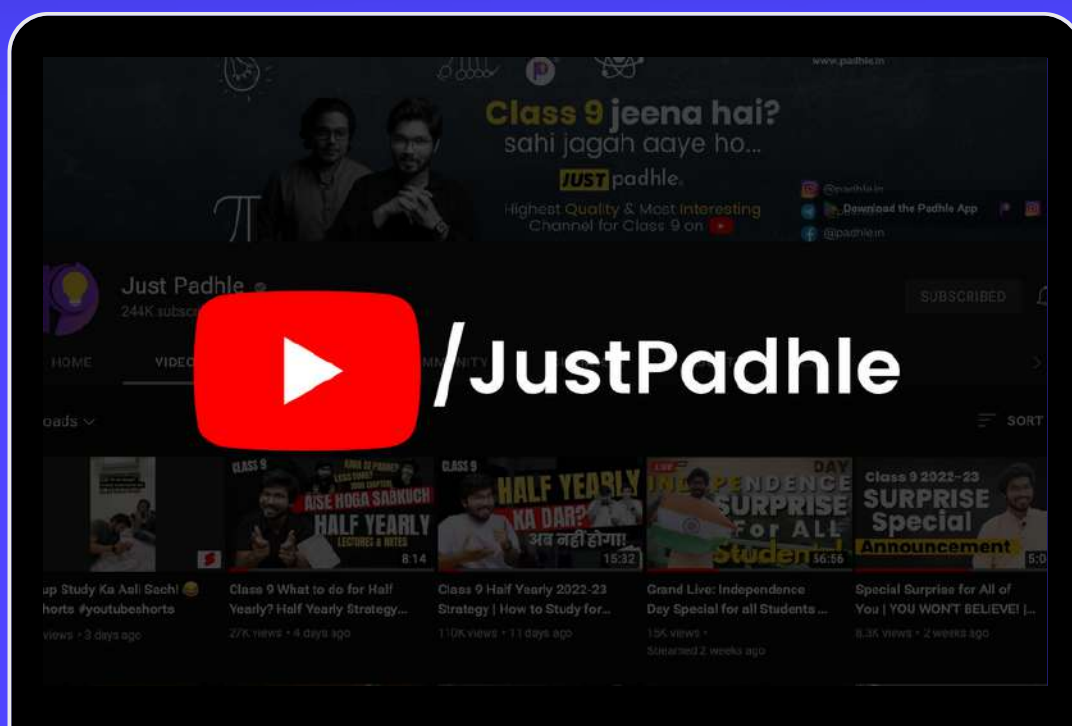
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Forces and Laws of Motion (जीवन के कानून)

* This is a vast chapter, but we will go pin point!

What is force?

→ The push or pull on an object with mass that causes it to change its velocity.

How to identify a force if applied, OR what does a Force change?

- Speed
- Direction
- Shape

- Force has both magnitude and direction, making it a vector quantity.

- It is measured in the SI unit of Newton (N).

- It is represented by the symbol F .

BALANCED

If the resultant of applied forces is equal to zero, it is called balanced forces.

For example : In the tug of war game when the force applied by both teams is equal in magnitude then the rope does not move in either side.

- This is due to the balanced forces in which resultant of applied forces comes out to be 0.

- Do not cause any change of state of an object.
- Are equal in magnitude and opposite in direction.
- Balanced forces can change the shape and size of an object.

Ex: When we press a balloon from opposite sides, the size of balloon is changed.

UNBALANCED FORCES

If the resultant of applied forces are greater than zero, are called unbalanced forces.

- To move an object unbalanced forces are to be applied from the opposite.
- In case of unbalanced forces acting on a body, it moves in the direction of the greater force.

* Unbalanced forces can:

- Change the speed and position of an object.
- Change the shape and size of an object.

CONTACT FORCES

- ♦ A force applied to an object by touching it physically.
- ♦ Example: holding, lifting etc.

NON-CONTACT FORCE

- ♦ A force applied to an object without touching it physically.
- ♦ Example: Magnetic force, Force of Gravitation

NEWTON'S LAWS OF MOTION (जीवन का सत्य)

* The easiest trick to remember all the laws of motion is mentioned in the video.

NEWTON'S FIRST LAW

⇒ An object remains in a state of rest or of uniform motion in a straight line unless compelled to change that state by an applied force.

⇒ The tendency of undisturbed objects to stay at rest or to keep moving with the same velocity is called inertia.

Inertia का easy मतलब —

- A body at rest tends to remain at rest.
- A body in motion tends to remain in motion.

* Example: When we sleep, we just want to keep sleeping, until mummy or papa scold us, that is they apply external force.

* But when we are with our friends outside enjoying/ partying, we want to stay there and don't come back home for resting, but we do have to, because our parents will apply external force and scold us.

(ये example सिर्फ समझाने के लिए है, exam में मत लिख आना !)

WHAT IS MOMENTUM?

* Momentum (p) is represented by Product of mass and velocity.

$$p = mv$$

Unit - kg ms^{-1}

NEWTON'S SECOND LAW ($F=ma$) -

* The rate of change of momentum of an object is proportional to the applied unbalanced force in the direction of force.

→ Let initial momentum before applying force = p_i
Let final momentum after applying force = p_f

According to this law,

$$F \propto \frac{p_f - p_i}{t}$$

$$\Rightarrow F \propto \frac{mv - mu}{t}$$

$$\Rightarrow F \propto m \left(\frac{v - u}{t} \right)$$

$$\Rightarrow F \propto ma$$

Applying constant 'k' to remove proportionality sign,

$$\Rightarrow F = kma$$

Experimentally it was verified that $k=1$.

$$\Rightarrow F = ma$$

WHAT IS IMPULSE?

⇨ Whenever a cricketer wants to catch a ball, he pulls the hands behind so that the time of contact increases and he feels less force.

$$F = ma$$

$$F = m\left(\frac{v-u}{t}\right)$$

→ That is,

'F' is inversely proportional to 't' that is time.

- More the time, less the force felt by the fielder.

→ On the other hand, if we catch a ball without proper technique, we can hurt ourselves because then, the time is very less.

- Hence force we feel is quite high.

NEWTON'S THIRD LAW (LAW OF KARMA)

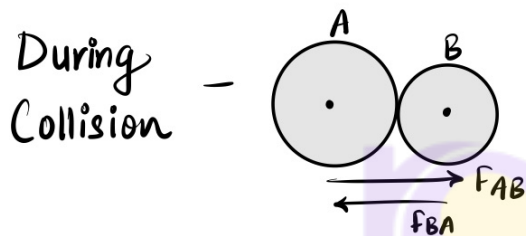
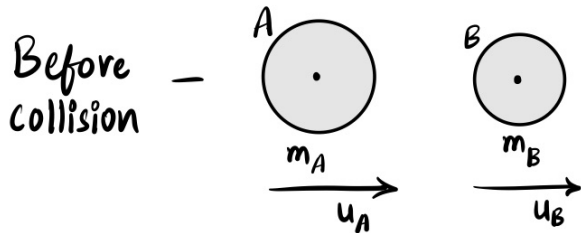
⇨ Every action has an equal and opposite reaction.

- These two forces are always equal in magnitude but opposite.
- These forces act on different objects and never on the same object.
- There is a pair of forces and not just one force.
- The two opposing forces are also known as action and reaction forces.
- A single force never acts.

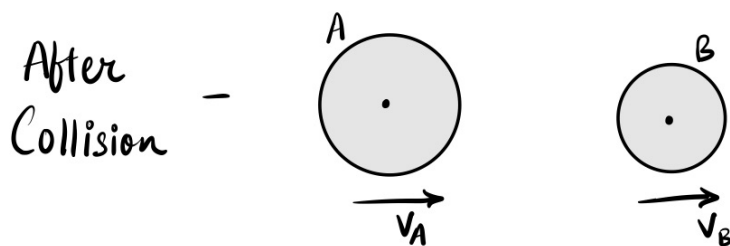
LAW OF CONSERVATION OF MOMENTUM (Very important, gets directly asked)

→ Consider 2 objects A and B, having mass m_A and m_B respectively.

- Let initial velocity of A be u_A and that of B be u_B .



Let force applied by A on B be F_{AB} ,
Let force applied by B on A be F_{BA} .



Let velocity of A and B after collision be v_A and v_B respectively.

→ Now, we know that according to Newton's 3rd law of motion, every action has an equal and opposite reaction.

⇒ During collision,

$$F_{AB} = -F_{BA}$$

$$\Rightarrow m_B a_B = -m_A a_A \left[\text{where } a_A \text{ and } a_B \text{ are accelerations of } A \text{ and } B \right]$$

$$= m_B \left(\frac{v_B - u_B}{t} \right) = -m_A \left(\frac{v_A - u_A}{t} \right)$$

$$= m_B v_B - m_B u_B = -m_A v_A + m_A u_A$$

$$= m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

$\Rightarrow$ Total momentum before collision =
Total momentum after collision.

Hence, proved!

$\rightarrow$ That means, momentum can neither be created nor be destroyed.
- In an isolated system, momentum always remains conserved.

Application of Law of conservation of momentum -

1. Recoil of gun
2. Collisions

1. Recoil of gun:

$\rightarrow$ The total momentum of the system of bullet and the gun before firing the bullet and after firing will be same.

$\rightarrow$ We can equate both the momentums and can find velocity of the gun, or the bullet whichever is asked for.

$$m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2$$